

Analyzing Distributions:

With distribution analysis in statistics, the question is rarely “does my data follow a normal distribution,” as this is very rare in the real world. The question is instead “how far does my data deviate from a normal distribution and what impact might that have on further tests?” Many statistical tests assume a normal distribution, and it is important to know the distribution of the data before applying them. At the same time, with large sample sizes, some statistical tests that would otherwise assume normal distributions (t-tests, ANOVA, regression) do not require this assumption to be met due to the Central Limit Theorem.

Continuous Data

The recommendation by statisticians is to plot the data, either by a histogram or a QQ plot, and visually determine whether the data follows a certain distribution.

If this is not possible or you have a large number of features, the Shapiro-Wilk, Anderson-Darling, and Kolmogorov-Smirnov tests are commonly used to analyze distributions. The Shapiro-Wilk test has the highest power (accuracy) but should only be used to analyze whether a data set with less than 50 samples follows a normal distribution. The Anderson-Darling test has the next highest power and should be used to analyze normal (for sample sizes > 50), exponential, and logistic distributions in Python. All other distributions can be analyzed with the K-S test.

Keep in mind that a numerical test may not be accurate at low sample sizes, as data sampled may not reflect the underlying distribution. A numerical test may also overemphasize outliers at large sample sizes: a slight deviation from normal can be enough for the test to classify the whole distribution as non-normal. However, that distribution may be “normal enough” for further statistical tests (see Statistical Tests section for more information).

The null hypothesis is that the data does follow the tested distribution: a p-value greater than .05 signifies that the data follows the tested distribution.

Other common statistics for exploratory data analysis include the mean, median, range, variance, spread, standard deviation, skew, kurtosis, and outliers.

Data can be positively or negatively skewed and leptokurtic or platykurtic. If a skew has an absolute value greater than .8 or kurtosis greater than 3 that is another sign the data is not drawn from a normal distribution.

Categorical Data

Common statistics for categorical data include the number of modes (unimodal/multimodal), number of categories, number of counts per category, and the distribution of the count data. This distribution analysis is best for ordinal or ranked categories as otherwise the order of the categories might change the underlying distribution (if you have bins A, B, and C with 10, 20, and 10 count each, running a distribution analysis on order ABC would result in a different outcome than BCA).

Once we know if the data is drawn from a normal distribution or not, we can then decide what kind of test we'd like to use.

Statistical Tests:

There are several different types of statistical tests that may be useful depending on user questions. All tests have a parametric (for when the data follows a normal distribution) and non-parametric (for when the data does not follow a normal distribution) version. **When the sample size is large enough (>50 data points), t-tests, ANOVA, and regression may be used regardless of the normality of the underlying distribution.** The null hypothesis with all these tests is that the data does not differ significantly: a p-value less than .05 signifies that the data does differ significantly.

Do two features have equal variance? Parametric version: F-test. Non-parametric version: Levene's test

Does a feature from a specific observation differ significantly from year 1 to year 2? Parametric version: Paired-test (used for time-series data or data that is related in some way). Non-parametric version: Wilcoxon signed rank test.

Does a feature from two different observations differ significantly? Parametric version: One-sample t-test. Non-parametric version: One-sample Wilcoxon signed rank test.

Does a feature from 3+ observations differ significantly? Parametric version: ANOVA. Non-parametric version: Kruskal-Wallis.

If we have a small sample (<20 observations), it is recommended that you use a non-parametric test.

Outliers:

Isolation Forest is a tree-based package in scikit-learn that can be used to detect outliers. In contrast to methods like IQR, it looks at multiple features in determining anomalies.

When doing exploratory data analysis, you can put all features (including the target variable) into Isolation Forest. When you're running a regression model, I would first perform a train-test split, then use Isolation Forest when finding outliers in my features and IQR when looking for outliers in my target variable.

Correlations:

When you have continuous features and a continuous target variable, you can run Pearson's Correlation.

When you have categorical features and a categorical target variable, you can run a Chi-Squared test. If the p-value is less than .05, you can reject the null hypothesis and it is likely that there is a statistically significant relationship between the two variables. You can follow up with a Cramer's V test to measure the strength of this relationship.

When you have categorical features and continuous target variable or categorical target variable and continuous features, you can run a one-way ANOVA test.

Multivariate Regression:

It's important to note that linear and logistic regression models have assumptions about the linearity of features and target variables that we aren't explicitly checking for in this analysis. It's possible that the data does not actually follow a linear relationship, in which case this type of

model would not be useful. I primarily focused on regression because of its coefficient interpretability. However, if the user is purely interested in predictive accuracy, a model like CatBoost (described below) would be a better general-purpose solution.

A regression model can be useful when we want coefficient interpretability (i.e. when our target variable increases by 1 unit, how much does each feature increase/decrease by?) Other machine learning models do not have this direct interpretability. Coefficients can also help us determine what features are most impactful for the prediction based on their absolute value.

Continuous target variable

There are several different types of regression for different use cases. All of them can be accessed through scikit-learn.

- “Vanilla” Linear Regression
 - o Works the best when all its assumptions are met!
 - o Susceptible to outliers
 - o Susceptible to multicollinearity
- Randomized Lasso – *regularized*
 - o Keeps only a subset of features
 - o Robust against outliers
 - o Handles multicollinearity
- Ridge Regression – *regularized*
 - o Keeps all features
 - o Susceptible to outliers
 - o Handles multicollinearity
- Elastic Net – *regularized*
 - o Best of both worlds between ridge regression and lasso regression
 - o Keeps only a subset of features
 - o Robust against outliers
 - o Handles multicollinearity
- Huber Regression

Some important considerations are:

- Outliers. Most linear regression models do best without outliers (all except Huber Regression); however, outliers can often be significant when interpreting the model. For example, a country with an extremely large GDP might be an outlier, but it could tell us something valuable about high-GDP countries. For this reason, statisticians usually do not recommend removing outliers. However, removing them significantly improves the performance of the model. One option would be to run the same model twice, once with outliers and once without, and show both results to the user so they can compare.
- Multicollinearity: Vanilla Linear Regression and Huber Regression both struggle when their features are highly multicollinear. One option is to check for multicollinearity using VIF before using these models. You can iteratively remove features with a VIF score over 10 (signifying high multicollinearity) until all features have a low enough score. This must be an iterative process as removing one feature may have an impact on the rest. Alternatively, you

can use a regularized model (Lasso, Ridge, Elastic Net) as they are designed to handle multicollinearity.

- Another way to remove multicollinearity is through PCA. Because this makes the variables less interpretable, I chose not to include this in my regression workflow. However, I include a section about it below.
- $N \ll P$: If the number of features is greater than the number of observations, you must use regularized linear regression.

Some questions that might help us determine what model to use are as follows:

Do we want to keep the outliers? Use Huber with VIF.

I don't mean this in the case of the side-by-side, outliers and no outliers model explained above, but in the case of when the user 100% wants to keep the outliers in the model but also wants the highest model accuracy.

Are we worried about multicollinearity but want to include all features in our prediction? Use Ridge Regression (note that it is susceptible to outliers).

Are we worried about multicollinearity but okay with dropping some features? Elastic Net.

Do we only want to select a subset of the most predictive features? Randomized Lasso.

Do we have more predictors than observations? Elastic Net.

As a general-purpose model, I would suggest **Elastic Net** as it is a mix of Lasso and Ridge regression and can "become" one or the other based on what leads to the highest accuracy. It does require the tuning of two hyperparameters.

Categorical Target Variable

The general LogisticRegressionCV is not susceptible to outliers and is a good multipurpose model.

Encoding Categorical Variables:

The best method for encoding categorical variables for use in a linear regression model is by one-hot encoding them. If possible, I would recommend one-hot encoding. However, this method can easily blow up the size of the data frame depending on the number of categorical variables and values.

An alternative method is target encoding, which assigns each categorical variable a value based on the estimate of average target values. Because this can introduce target leakage, I would recommend using CatBoost encoder (a variant of target encoding that prevents target leakage). CatBoost encoder works for continuous and binomial target variables. For multiclass target variables, the general-purpose Target Encoder is needed.

Linear Regression Workflow:

- Train/test split – always do this first to prevent target leakage!

- Encode categorical variables
- Remove columns with 0 variance
- Check for outliers using Isolation Forest and IQR
- Check for multicollinearity using VIF
- Scale data with StandardScaler
- Determine what type of regression model we plan to use (Randomized Lasso, Elastic Net, Ridge Regression, vanilla OLS) based on user interests
 - o If dataset greater than 100K rows, use SGDRegressor
- Run model
- If Durbin-Watson returns a high chance of autocorrelation, suggest a time series model.

Dimensionality Reduction:

PCA is a method for dimensionality reduction that also eliminates multicollinearity. Data should be scaled before running PCA, and PCA should be performed before running models like K-means clustering that are susceptible to the curse of dimensionality (a high feature space).

Determining the ideal number of PCs is best done through visual inspection. Plotting a scree plot and looking for the “elbow” can tell you at what point adding additional PCs has little to no impact. Another option is to look for the point at which the variance explained by the PCs is greater than 95%.

You can analyze what variables make up each PC by looking at the loadings. This will tell you how much each variable contributes to the PC. Large loadings signify a large contribution.

Other Predictive Models:

CatBoost:

CatBoost is a tree-based model that is not impacted by scale or multicollinearity; it also does not require categorical variables to be encoded. However, it can be susceptible to outliers.

The predictive accuracy of CatBoost is comparable to other tree-based models such as XGBoost. Its main benefit in our case is that we can just throw all our variables into the model without worrying about encoding/scaling/etc.

It's important to note that tree-based models are non-parametrized and so we cannot interpret the coefficients as we did with our regression models. We can use feature importance scores to analyze the contribution of each feature towards the overall prediction, with larger scores meaning a larger contribution, but we cannot determine precisely how much each coefficient varies with regards to the target variable.

We can analyze the contribution of each feature (and sub-group within a categorical feature) through SHAP plots and values.

K-means clustering:

After running PCA on your data, you can use K-means as an exploratory tool to group your observations. Keep in mind that K-means is not a stable model and repeated iterations will lead to different clusterings. Categorical variables must be encoded before running K-means.

An elbow plot can once again be used to determine the optimal number of clusters. Adding cluster labels as a new feature can be used as a tool to do additional descriptive statistics within clusters.

ARIMA (time-series analysis):

The simplest model is simply `auto_arima`, which will perform a grid search to set the three ARIMA hyperparameters for you (this can be computationally expensive). If there is a stationarity issue, an error would be thrown, suggesting that differencing the data to achieve stationarity might be needed. Once this model is built, it can also be used to forecast for the future.

You will also have to determine if the data is seasonal before using the model. This can best be done through visual inspection (is there a repeated spike at a certain time of year/every few years?) and specify the period of seasonality (the length of the cycle after which your data repeats).